

P8B Profiling: Qwen3-8B-Base on ASearcher (32-turn, top-5, 16k) vanilla vs. synchronous partial rollout, GRPO vs. turnPPO

Agentic-RL-CA rung-3 — branch `sync-partial-rollout-poc`

2026-07-29/30 (SF time)

TL;DR

All four configs run end-to-end on one 8×H100 worker. The partial-rollout PoC is **functionally correct** (T1/T3-lite resume identity byte-exact; group-atomic release invariants hold in every cycle; zero stale drops) and gives **1.5–1.6×** overall throughput (2.2–2.5× on generation). However, **no configuration meets the ~3-day/arm gate at the full protocol budget**: projected 150-step wall-clock is 13.3 d (GRPO vanilla) / 8.9 d (GRPO partial) / 20.3 d (turnPPO vanilla, pre-fix estimate) / 12.8 d (turnPPO partial). Decision options in §5; **no arm was auto-launched**.

1 Setup

Qwen3-8B-Base, RL-from-base, ASearcher Base-35k *unfiltered*, wiki-18/e5 top-5, one 5k-char `<information>` block per turn (never discarded), full `<think>+<search>` responses retained in history, 32-turn nominal horizon, 16k prompt with *clean context-overflow termination*, 1024-token responses, $256 \times 5 = 1280$ trajectories/step, MICRO_BSZ=1. Partial rollout: `cycle_turns=8`, `max_age=4`, uid-group-atomic release, no off-policy correction (per-row `policy_version` recorded). Profiling: 2 steps/config vanilla, 4 cycles/config partial (turnPPO-partial cycle 4 cut by the 6 h per-config timeout; 3 clean cycles suffice). Runs: `p8bprof_*` in wandb project [ca-rung3-8b-asearcher](#).

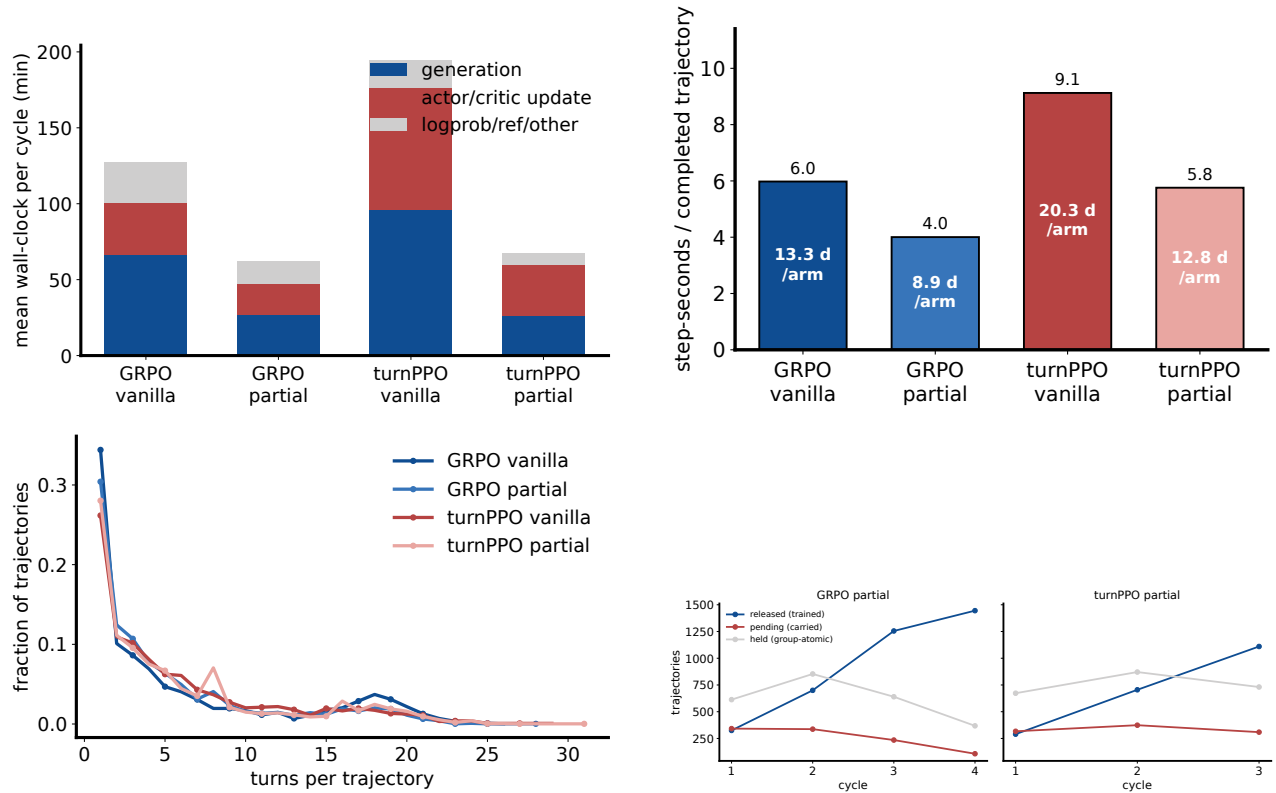
2 Correctness of the partial-rollout PoC

- **T1 resume identity** (scripted actions, live retriever): prompts restored from snapshots byte-identical to the uninterrupted reference, before *and* after a post-restore step (done-flags included). **T3-lite**: `reset_partial([])` $\equiv$ vanilla reset. Log: `t1_resume_identity.log`.
- **Conservation invariants** hold exactly in all 7 partial cycles observed: `carried + fresh = released + pending + held`; released always divisible by group size 5.
- **Staleness**: mean open-group age 0.35–0.45 cycles; `dropped_stale_groups=0` throughout (≤ 1 policy-version lag for most resumed segments).

3 Throughput and memory

	GRPO vanilla	GRPO partial	turnPPO vanilla [†]	turnPPO partial
step/cycle wall-clock (mean, s)	7652	3728	11681	4041
generation share (s)	3961	1591	5740	1585
update share (s)	2086	1262	4850	2009
step-seconds / completed traj	5.98	4.00	9.13	5.76
projected 150-step arm (days)	13.3	8.9	20.3	12.8
peak mem alloc (GB)	85.2	70–85	OOM@KV0.6 [†]	fits @KV0.5+offload

[†]turnPPO vanilla crashed at cycle 2’s vLLM `wake_up(kv_cache)` under KV fraction 0.6 with the colocated 8B critic; its numbers above are from its single completed step. A retry under the fixed posture (KV 0.5 + critic CPU offload, matching turnPPO partial) is in flight; expect update share to grow somewhat from offload.



Generation is the straggler-bound share and is where partial rollout pays: 2.2–2.5 $\times$. Updates scale with trained tokens (identical either way), diluting overall gain to 1.5–1.6 $\times$. Buffer dynamics (bottom-right) show steady state after cycle 2: $\sim$ 1250–1450 released/cycle at 1280-slot pool, pending 100–370.

4 Trajectory behaviour (RL-from-base sanity)

- **Grammar compliance out of the box:** step-0 greedy val em **0.199** (512 unfiltered questions), avg 8.2 turns, 7.6 searches/traj, 0% unterminated — the base 8B uses the tool grammar zero-shot (the 4B-Instruct greedy baseline was 0.010). Training valid-action ratio 0.315 at step 1 (temp 1.0), entropy 2.57 $\rightarrow$ 1.97 by step 2.
- **Length wall:** mean 5.3–6.1 turns, max 26 (<32): the 16k prompt binds first, as designed. Ctx-terminate produced clean endings; residual prompt-clip 0.5% (char prefilter tail — tighten threshold before headline runs).
- **Caps:** 5k info-block cap hits 3–8% of retrievals; 1024-token response cap hits 24–37% of turns (long base-model thinks).
- **Train success** 0.16–0.20 at temp 1.0 (unfiltered data; parametric shortcut present by design — cf. the filtered-4B step-0 of 0.008).

5 Decision needed (no auto-launch)

The grilled go-criterion “ $<\sim 3$ days/arm” fails for every config, so the two 150-step arms were **not** launched. Options, combinable:

1. **Partial rollout for both arms** (8.9d GRPO / 12.8d turnPPO): systems-faithful to ASearcher, but couples the algorithm comparison to the PoC’s uncorrected staleness (R2).
2. **Cut steps to 75** (≈ 4.5 d / 6.4d with partial): halves optimization budget; 4B curves suggest most movement happens by step 75.
3. **Shrink batch or group** (e.g. 128 $\times$ 5): halves step cost, halves trajectories/update; breaks strict budget-matching with the 4B arms.

4. **Accept the wall-clock** for vanilla arms (13–20 d each) — cleanest science, slowest calendar, occupies 2 of ~ 6 quota slots throughout.

Reproduce

`scripts/asearcher/p8b_profile_worker.sh` (worker; `ONLY_CFG` to subset), `analyze_profiling.py` + `make_figures.py` (this folder). Raw per-turn dumps: `HDFS logs/p8b_profile/rollout_*.jsonl.zst`. Design doc: `docs/design/2026-07-29_sync_`